

MEDTRACK HOSPITAL ANALYTICS

Project Report on

Hospital Admissions, Resource Utilization & Department Performance Dashboard

Tools & Technologies: Python (NumPy, Pandas) · Jupyter Notebook · Tableau
50,213 patient admissions · 10 departments · 6 global regions

Prepared by: Aryan Kavan Gowda

Table of Contents

Table of Contents	2
1. Executive Summary	3
2. Project Objective.....	3
3. Tools & Technologies Used	4
4. Methodology — Project Milestones	4
5. Milestone 1 — Data Collection & Cleaning.....	5
5.1 Data Collection.....	5
5.2 Cleaning & Transformation Steps	5
5.3 Validation Results.....	5
6. Milestone 2 — KPI Engineering.....	6
6.1 Dashboard Storyboard.....	6
7. Milestone 3 — Dashboard Build (Tableau).....	7
7.1 Dashboard 1 — Hospital Overview	7
7.2 Dashboard 2 — Department Efficiency.....	7
7.3 Supporting Worksheets.....	7
8. Milestone 4 — Testing & Validation	8
9. Key Insights	8
10. Conclusion	8
Appendix A — Data Dictionary (Core Fields)	9

1. Executive Summary

MedTrack is an end-to-end hospital analytics project that transforms raw, multi-table hospital operations data into a validated, analysis-ready dataset and a set of interactive Tableau dashboards. The project covers the full pipeline: data collection and cleaning, KPI engineering, dashboard design, and build, using Python for data preparation and Tableau for visualization.

The final dataset spans 50,213 patient admissions across 10 hospital departments and 6 global regions, covering admission and discharge tracking, bed and equipment utilization, staffing, treatment cost, insurance, and patient satisfaction.

Key result: the project delivers two production Tableau dashboards — Hospital Overview and Department Efficiency — built from 17 individual worksheets, backed by a cleaned and validated data pipeline with zero orphaned records.

2. Project Objective

The objective of this project is to give hospital administrators, department heads, and operations teams a single, interactive analytics layer to answer four core questions:

- ▶ How is the hospital performing overall, across admissions, occupancy, and length of stay?
- ▶ Where are the bottlenecks in the patient journey from admission to discharge?
- ▶ How does each department compare on efficiency, readmission, and patient satisfaction?
- ▶ Are beds, equipment, and staff being utilized efficiently across departments?

3. Tools & Technologies Used

Tool	Role in Project
Python (Pandas, NumPy)	Data loading, cleaning, type conversion, KPI calculation, and validation logic
Jupyter Notebook	Interactive environment for iterative cleaning, exploratory analysis, and KPI computation
Microsoft Excel (.xlsx)	Intermediate storage format for the cleaned dataset and pre-aggregated KPI summaries
Tableau Desktop	Dashboard design, worksheet construction, and interactive visualization

4. Methodology — Project Milestones

The project was executed across four milestones, moving from raw data to a fully interactive dashboard:

Milestone	Focus	Status
1. Data Collection & Cleaning	Consolidate source tables, validate integrity, standardize formats	Complete
2. KPI Engineering & Storyboard	Compute core metrics, plan dashboard layouts	Complete
3. Dashboard Build	Build interactive dashboards in Tableau	Complete
4. Testing & Delivery	Validate calculations, verify interactivity, finalize documentation	Complete

5. Milestone 1 — Data Collection & Cleaning

5.1 Data Collection

The source dataset (hospital final datasets.xlsx) was loaded using Pandas, with each admission record capturing patient, department, bed, staffing, equipment, billing, and outcome information in a single wide table (Cleaned_Data).

5.2 Cleaning & Transformation Steps

- ▶ Type correction: Admission_Date and Discharge_Date converted to native date types for time-based analysis.
- ▶ Derived fields: Length_of_Stay_Days computed and validated against admission/discharge date pairs.
- ▶ Standardization: categorical fields (Department, Admission_Type, Discharge_Status, Insurance_Type) checked for consistent casing and spelling.
- ▶ Null handling: fields with structurally valid nulls (e.g. charge references that only apply to certain charge types) were flagged rather than dropped, preserving row counts.
- ▶ Referential validation: every foreign-key-style relationship (patient → department → ward → bed) was checked to confirm zero orphaned records before proceeding to analysis.

5.3 Validation Results

Metric	Result
Total validated admissions	50,213
Departments	10 (Oncology, Pediatrics, Gynecology, Neurology, Orthopedics, ICU, Emergency, Surgery, Cardiology, General Medicine)
Regions	6 (North America, South America, Asia, Africa, Europe, Oceania)
Duplicate rows	0
Orphaned foreign keys	0

6. Milestone 2 — KPI Engineering

Nine core KPIs were engineered from the cleaned dataset using Pandas aggregation logic, later reproduced as native calculated measures for the Tableau layer. All figures below are computed directly from the validated 50,213-row dataset.

KPI	Formula	Result
Total Admissions	COUNT(admission records)	50,213
Average Length of Stay	AVG(Discharge_Date – Admission_Date)	4.42 days
Readmission Rate	% of admissions flagged Readmission_Flag = Yes	12.01%
Average Occupancy Rate	AVG(Occupancy_Rate) across departments	69.96%
Average Bed Utilization Rate	AVG(Bed_Utilization_Rate_Pct)	69.96%
Average Treatment Cost	AVG(Treatment_Cost_USD)	\$3,736
Average Patient Satisfaction	AVG(Patient_Satisfaction_Score), 1–5 scale	3.00
Deceased Outcome Rate	% of admissions with Discharge_Status = Deceased	1.92%
Emergency Admission Rate	% of admissions with Admission_Type = Emergency	34.93%

6.1 Dashboard Storyboard

Before building in Tableau, each dashboard was planned against a specific audience and core question, with chart types selected deliberately for the data being shown (trend lines for time series, horizontal bars for department comparisons, donuts only for low-cardinality breakdowns).

7. Milestone 3 — Dashboard Build (Tableau)

The final Tableau workbook connects to Cleaned_Data as the primary data source, with Department_KPIs and Overall_KPIs used as supporting pre-aggregated tables. Two dashboards were built, consolidating 17 individual worksheets:

7.1 Dashboard 1 — Hospital Overview

A 12-sheet executive dashboard answering “How is the hospital performing overall?”

Component worksheets:

- ▶ Total Admission, Occupancy Rate, Average Length of Stay, Bed Utilization Rate, Readmission Rate, Department Efficiency Rate — headline KPI views
- ▶ Monthly Admissions vs Discharges — patient flow trend over time
- ▶ Admissions Trend — overall admissions trend line
- ▶ Admission by Department, Admission by Patient Type — categorical breakdowns
- ▶ Average Length of Stay by Department, Readmission Rate by Department — department-level comparisons

7.2 Dashboard 2 — Department Efficiency

A focused 3-sheet dashboard for department-level comparison, answering “How does my department compare to others?” Component worksheets:

- ▶ Department Efficiency — composite efficiency score by department
- ▶ Bed Utilization Rate by Department — resource utilization comparison
- ▶ Occupancy Rate vs Total Beds — capacity vs. demand view

7.3 Supporting Worksheets

Two additional worksheets — Discharge Tracking and Patient Movement — were built as supporting analysis views alongside the two published dashboards, available for further drill-down.

8. Milestone 4 — Testing & Validation

- ▶ Every KPI value shown on the dashboards was cross-checked against the equivalent Pandas aggregation from the cleaning notebook.
- ▶ Referential integrity between Cleaned_Data and the Department_KPIs summary table was verified on the Department field before publishing.
- ▶ All worksheets were visually reviewed for correct chart type, axis scaling, and label formatting prior to dashboard assembly.

9. Key Insights

- ▶ Average length of stay across the hospital is 4.42 days, with meaningful variation across departments — the Department Efficiency dashboard isolates which departments run longer or shorter than average.
- ▶ A readmission rate of 12.01% provides a clear baseline for tracking care-quality initiatives over time.
- ▶ Occupancy and bed utilization both average just under 70%, suggesting headroom in most departments alongside likely pockets of over-capacity worth investigating department-by-department.
- ▶ Just over a third of all admissions (34.93%) enter through Emergency, underscoring the importance of the Patient Flow-style views for operational planning.

10. Conclusion

This project demonstrates a complete analytics workflow: from a raw, multi-attribute hospital operations dataset, through Python-based cleaning and validation, to a set of interactive Tableau dashboards that surface admissions, resource utilization, and department performance in a single, navigable interface. The resulting Hospital Overview and Department Efficiency dashboards give stakeholders a validated, real-time view of hospital operations without needing to touch the underlying data directly.

Appendix A — Data Dictionary (Core Fields)

Field	Description
Patient_ID / Hospital_ID / Doctor_ID / Bed_ID	Unique identifiers for patient, hospital, doctor, and bed
Department / Region / Hospital_Name	Organizational and geographic dimensions
Admission_Date / Discharge_Date / Length_of_Stay_Days	Core time-based admission fields
Admission_Type / Patient_Type / Discharge_Status	Categorical classification of the admission and its outcome
Readmission_Flag	Yes/No flag for readmission within the tracked window
Total_Beds_In_Department / Occupied_Beds_At_Admission / Occupancy_Rate / Bed_Utilization_Rate_Pct	Bed capacity and utilization fields
Staff_On_Duty / Equipment_Used / Equipment_Utilization_Rate_Pct	Staffing and equipment utilization fields
Treatment_Cost_USD / Insurance_Type	Financial fields
Patient_Satisfaction_Score / Department_Efficiency_Score	Outcome and performance scoring fields